

Isabel Panicker

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Education

Georgia Institute of Technology, School of Chemical and Biomolecular Engineering, Atlanta, GA August 2023 - Current

PhD Candidate, Chemical and Biomolecular Engineering

Minor in Materials Science and Engineering

Focus: liquid-phase transmission electron microscopy, single-particle tracking, nanotechnology, soft matter

Advisor: Dr. Vida Jamali

Committee: Dr. Saad Bhamla (ChBE), Dr. David Flaherty (ChBE), Dr. Joshua Kacher (MSE), Dr. Hang Lu (ChBE)

Purdue University, Davidson School of Chemical Engineering, West Lafayette, IN

May 2023

B.S. Chemical Engineering, Minor in Management

Research Experience

PhD Candidate - Jamali Lab | Georgia Institute of Technology September 2023 - Current

- *Thesis:* Developing liquid-phase transmission electron microscopy as a robust single-particle tracking tool to probe nanoscale transport and complex mechanics of soft matter systems.
- Advanced LPTEM as a single-particle tracking (SPT) platform by showcasing how ionic charge environment modulates liquid cell interfacial chemistry that generates non-Brownian nanoparticle dynamics.
- Applied canonical statistical diffusion analysis and deep-learning-based diffusion classification on particle trajectories to reveal environment-specific transitions of diffusion mechanisms.
- Developed the first passive nanorheology framework in LPTEM to model these dynamics, enabling quantitative extraction of complex viscoelastic moduli from both translation and rotational nanoparticle trajectories.
- Built SAM-EM, a custom Python-based tool to enhance particle tracking precision and acquisition throughput with LPTEM, finetuning Meta's Segment Anything Model 2 (SAM-2) on realistic, simulated LPTEM data, and integrated with particle tracking and statistical analysis tools, enabling zero-shot segmentation of noisy LPTEM acquisitions.
- Revealed the active motion and the resulting compositional polarity dependency of ~100nm catalytic platinum-polystyrene Janus nanomotors utilizing a correlative microscopy platform with TIRF and LPTEM, demonstrating how catalytic asymmetry can continue to bias displacement direction despite thermal fluctuations unique to the nanoscale.
- Utilizing the LPTEM-based passive nanorheology framework to map nanoscale viscoelasticity in liquid-phase separated systems, revealing how spatially heterogeneous mechanics evolve at the nanoscale and characterizing the phase behaviors of nanoscale phases.
- Performed LPTEM imaging of Self-Agglomerating Nanohydrogels (SANGs) to visualize structural evolution under variable environmental conditions as part of a collaborative drug delivery study.
- Mentoring undergraduate researchers in laboratory skills, nanomaterial synthesis, and microscopy, enabling them to carry out independent experiments and contribute meaningfully to research projects.

Undergraduate Researcher - Solar Energy Group | Purdue University

January 2022 - June 2023

- Selected as a First-Time Researcher Fellow, a competitive program supporting initial undergraduate involvement in high-impact research.
- Developed and optimized solution-based synthesis protocols for multinary sulfide thin films (e.g., $\text{Cu}_2\text{BaSnS}_4$, alkaline earth metal chalcogenide perovskites) as energy efficient, earth abundant alternatives for photovoltaic applications.
- Synthesized crystalline sulfide materials via low-temperature decomposition of thiol-metal precursors, systematically screening thiol-metal combinations across solvent environments and confirming phase purity and composition using Raman spectroscopy, XRD, and XRF.
- Completed a senior thesis titled "*Solution Based Synthesis of Alkaline Earth Metal Thiolate Precursors for Photovoltaically Active Multinary Sulfides.*"

Professional Experience

Drive Unit Quality Engineering Intern | Tesla Gigafactory, Reno, Nevada

May 2022 - August 2022

- Utilized SolidWorks to create a 3D printed tool to aid rundown in the manufacturing process.
- Wrote and published manufacturing instructions on individual Drive Unit part storage prior to workstation introduction.
- Cross-functionally managed the acquisition and installation of a water in oil sensor utilized inline.
- Investigated part contamination resulting from current storage practices by designing and subsequently conducting cleanliness tests.
- Executed validation trials to ensure supplier compliance to dimension specs and prevent part starvation.

Publications

Peer-Reviewed Articles

- **Panicker, I.**, Shabeeb, Z., Hargus, C., Jamali, V. "Solution-Tunable Interfacial Interaction Landscape Governs Anomalous Nanoparticle Diffusion in Liquid-Phase Electron Microscopy." *ACS Nano* (2026) (Featured on Front Cover).
- Shabeeb, Z., Sreeram, S., **Panicker, I.**, Wang, A., Xu, M., Goel, R., Jamali, V. "SAM-EM: Segment Anything for Liquid-Phase Electron Microscopy Video Segmentation." *Digital Discovery* (2026).
- Housley, S.N., Bourque, A.R., Matyunina, L.V., **Panicker, I.**, Echeverri, E.S., Izykiewicz, M., Herrmann, O.A., Lee, S., Arboleda, J.C., Jamali, V., McDonald, J.F., Dahlman, J.E., Finn, MG. "Tumor Agnostic Drug Delivery with Self-Agglomerating Nanohydrogels (SANGs)." *Nature Communications* (2026).
- Agarwal, S., Vincent, K. C., Turnley, J. W., Hayes, D. C., Uible, M. C., Durán, I., Canizales, A. S. M., Khandelwal, S., **Panicker, I.**, Andoh, Z., Spilker, R. M., Ma, Q., Huang, L., Hwang, S., Kisslinger, K., Svatek, S., Antolin, E., Bart, S. C., & Agrawal, R. "Breaking Barriers in Chalcogenide Perovskite Synthesis: A Generalized Framework for Fabrication of BaMS₃ (M=Ti, Zr, Hf) Materials." *Advanced Functional Materials* (2024).
- Turnley, J. W., Vincent, K. C., Pradhan, A. A., **Panicker, I.**, Swope, R., Uible, M. C., Bart, S. C., & Agrawal, R. (2022). "Solution Deposition for Chalcogenide Perovskites: A Low-Temperature Route to BaMS₃ Materials (M = Ti, Zr, Hf)." *Journal of the American Chemical Society* (2022).

Conference Proceedings/Workshop Papers

- Wang, A., Xu, M., Goel, R., Shabeeb, Z., **Panicker, I.**, Jamali, V. "SAM-EM: Real-Time Segmentation for Automated Liquid Phase Transmission Electron Microscopy." *AI for Accelerated Materials Design - NeurIPS* (2025).

Manuscripts Under Review and Pre-prints

- **Panicker, I.**, Roque, K.I., Ruiz, R., Ben-Moshe, A., Bustamante, C., Alivisatos, A.P., Jamali, V. "Polarity-resolved Imaging Reveals Active Transport in Catalytic Janus Nanomotors." *ChemRxiv* (2026).

Presentations and Panels

- "Revealing Active Transport of Catalytic Janus Nanomotors by Polarity-Resolved Imaging" - GT ChBE 4th Year Colloquium Fall 2026
- "Revealing Superdiffusive Active Motion of Nanoscale Particles with Correlative Light and Electron Microscopy" - ACS Spring 2026
- "Programming and Probing Interfacial Transport in Liquid-Phase TEM through Ion Screening and Passive Nanorheology" - ACS Spring 2026
- "Programming and Probing Interfacial Transport in Liquid-Phase TEM through Ion Screening and Passive Nanorheology" - Invited Talk at the 2026 Gordon Research Seminar on Liquid-Phase TEM (Spring 2026)
- "Screening Nanoparticle Surface Interactions at the Liquid-Solid Interface in Liquid-Phase Transmission Electron Microscopy" - Poster Presentation at Soft Matter Day (Spring 2025)
- "Developing a Single Particle Tracking Method through Correlative Microscopy" - Poster Presentation at American Chemical Society Colloid and Surface Science Symposium (Summer 2024)
- "Developing a Correlative Microscopy Framework for Characterizing Motion of Active Matter at the Nanoscale" - Poster Presentation at Soft Matter Day (Spring 2024)
- "Organizing a Successful Research Symposium" - Panelist at AIChE Annual Student Conference (Fall 2023)
- "Synthesizing Metal Thiolates as Soluble Precursors for Multinary Sulfides" - Poster Presentation at AIChE Annual Student Conference (Fall 2023); Poster Presentation at Purdue University AIChE Undergraduate Research Symposium (Spring 2023); Poster Presentation at Purdue University Undergraduate Research Conference (Spring 2023)
- "Soluble Precursors for Chalcogenide Semiconductors Containing Alkaline Earth Metals" - Poster Presentation at Purdue University Undergraduate Research Conference (Spring 2022)

Honors and Awards

Georgia Tech ChBE Eckert Graduate Fellow	Spring 2026
Shell Outstanding Teaching Assistant Award	Spring 2026
NSF Graduate Research Fellowship Program - Honorable Mention	Spring 2025
2023 AIChE Undergraduate Research Symposium 1 st Place Awardee - Purdue University	Spring 2023

2022 Spring Undergraduate Research Conference 2 nd Place CoE Awardee - Purdue University	Spring 2022
First-Time Researcher Fellowship, Purdue University	Fall 2021
Purdue University Dean's List	2019 - 2023
Purdue University Semester Honors	2019 - 2023

Press

Georgia Tech: Study Uncovers New Insights into Nanoparticle Motion (06/05/2026)
Medical Press: Nanohydrogels steer cancer drugs to tumors, aiming to spare healthy tissue (03/10/2026)
Georgia Tech: Target the Tumor. Spare the Body. (03/06/2026)

Leadership, Involvement, and Service to Scientific Community

National Nanotech. Coordinated Infrastructure Research for Teachers Mentor Georgia Tech	Summer 2026
ChBE Graduate Student Recruitment Chair Georgia Tech	Fall 2024 - Spring 2026
Graduate Student Safety Council - Inspections Officer Georgia Tech	January 2024 - Fall 2025
STEM Outreach Volunteer Georgia Tech	Spring 2024- Fall 2025
ChBE Mentor Georgia Tech	Fall 2024
Assistant Director TEDxPurdueU, Purdue University	April 2022 - April 2023
Peer Research Mentor Office of Undergraduate Research, Purdue University	Fall 2022
Chemical Engineering Ambassador Purdue University	April 2022 - May 2023
Logistics Co-Chair TEDxPurdueU, Purdue University	April 2021 - April 2022
Philanthropy Committee Member American Institute of Chemical Engineers, Purdue University	Fall 2020 - Spring 2021

Teaching Experience

Graduate Teaching Assistant: Unit Operations (ChBE 4200/4210)	Fall 2026
Graduate Teaching Assistant: Fluid Mechanics (ChBE 3205)	Fall 2024 and 2025

Additional Graduate Courses Taken

MSE 6110 (Transmission Electron Microscopy)	Spring 2024
MSE 8803 (Soft Nano and Biomaterials)	Spring 2025
PHYS 6205 (Soft Matter Physics)	Spring 2026
ChBE 8803 (AI for Chemical Engineers)	Fall 2026